

Edwin Lotsu

Clemson University, SC, USA | (703) 947-9499 | lotsuedwin@gmail.com | [LinkedIn- Edwin Lotsu](#) | [GitHub- Edwin Lotsu](#)

Professional Summary

Security-focused Computer Engineering graduate student with hands-on experience in SIEM deployment, endpoint detection engineering, and network security monitoring. Built and operates an enterprise-grade home lab running Wazuh SIEM with Active Directory, custom detection rules, and MITRE ATT&CK-mapped attack simulations. CompTIA Security+ certified. Experienced in access control, compliance workflows, and technical documentation in a real institutional environment.

Certifications

CompTIA Security + May 2026

Education

Clemson University, Clemson, SC Present

MSc, Computer Engineering

- **Expected Graduation:** December 2026
- **Relevant Coursework:** Distributed Denial of Services Attacks (DDOS), Computer Network Security, Malware Reverse Engineering

Ghana Communication Technology University, Ghana 2023

BSc, Computer Engineering

Experience

Enscript Solutions Ltd Nov 2023 - Dec 2024

Software Engineer

Accra, Ghana

- Designed and implemented a USSD-based fintech payment platform incorporating role-based access control, API token management, and multi-factor authentication across a 3-tier architecture.
- Applied input validation and sanitization controls across REST API endpoints to mitigate injection attack vectors and protect payment transaction integrity.
- Collaborated with project managers and QA teams to deliver client-facing fintech products on schedule.

Immigration Services, Clemson University Jan 2025 – May 2026

Graduate Assistant

Clemson, SC

- Administered identity and access management workflows in Sunopsis, enforcing least-privilege access controls and maintaining compliance with institutional data governance policies for sensitive student records.
- Developed technical documentation and training materials supporting onboarding and security compliance awareness for staff and international students.
- Managed institutional datasets using Excel and SharePoint, maintaining data integrity standards across reporting and IT operations.

Projects

Security Monitoring and Detection Engineering Home Lab

- Built a multi-VM security operations lab using Wazuh SIEM to monitor Windows Server (Active Directory), domain-joined Windows 10 endpoint, and Ubuntu infrastructure; deployed Sysmon for enhanced endpoint telemetry and simulated attack scenarios including network reconnaissance (Nmap port scanning) while analyzing detection artifacts through SIEM dashboards and structured queries.
- Developed custom detection rules to identify suspicious network activity patterns and privilege escalation attempts; mapped alerts to MITRE ATT&CK techniques including Network Service Scanning (T1046) and validated detection coverage through offensive simulations
- **Tools:** Wazuh, Sysmon, Kali Linux, Nmap, Windows Server (Active Directory), Ubuntu, VMware

Network Security & Malware Analysis – Academic Research Projects

- Built a comparative IDS pipeline using Snort and Suricata; evaluated detection rule effectiveness against evasion techniques and analyzed false positive rates using Wireshark traffic captures
- Performed static and dynamic analysis on malware using IDA Pro, Process Monitor, Process Explorer, and PE Viewer to identify attack vectors, persistence mechanisms, and indicators of compromise (IOCs).
- Simulated buffer overflow exploitation and packet sniffing; analyzed encrypted traffic behavior under VPN tunneling to assess detection visibility.
- **Tools:** IDA, Process Monitor, Virtual Private Networks, Wireshark, Linux, Snort, Suricata

Skills

- **Security Tools:** Wazuh SIEM, Sysmon, Snort, Suricata, Wireshark, IDA Pro, Nmap, Kali Linux
- **Detection & Analysis:** Custom Detection Rules, MITRE ATT&CK Mapping, Log Analysis, Threat Hunting, Malware Static & Dynamic Analysis
- **Infrastructure:** Windows Server (Active Directory), VMware, Ubuntu Server, Linux Administration
- **Programming:** Python, JavaScript